



Laboratoire
Méthodes
Formelles

école
normale
supérieure
paris–saclay

université
PARIS-SACLAY

RESEARCH PROJECT (TER) REPORT - SPRING 2026

Distributed framework for #SAT solvers

Thanh Phat Doan

Under the supervision of

Mihaela Sighireanu

Formal Methods Laboratory (LMF)

ENS Paris-Saclay

February 19, 2026

Préface

TODO

Abstract

TODO

Contents

A	Introduction	2
B	Project progression	3
B.1	Preliminary work	3
B.2	Phase 1: Design and initial implementation	3
B.3	Phase 2: Distributed Infrastructure and Networking	3
	Bibliography	4

CHAPTER A

Introduction

(TODO) the general context of the project

[1] [2] [3] [4] [5] [6]

CHAPTER B

Project progression

This chapter details the practical realization of the HydraSAT framework, documenting the transition from a conceptual design to a functional distributed system. We describe the project's evolution through several key development phases, highlighting the technical milestones achieved and the engineering decisions prompted by infrastructure constraints.

B.1 Preliminary work

Before embarking on the implementation of the framework, we conducted a comprehensive preliminary study to establish a rigorous theoretical foundation. Our investigation focused on the mechanics of modern SAT and #SAT solvers. Specifically, we analyzed the Conflict-Driven Clause Learning (CDCL) algorithm and compared chronological versus non-chronological backtracking strategies for solving propositional satisfiability problems [1]. Furthermore, we surveyed the current landscape of SAT methodology, including exact and approximate model counting techniques, component decomposition, and hashing-based approaches [2]. This theoretical grounding was essential for designing a distributed architecture capable of handling complex model-counting tasks.

B.2 Phase 1: Design and initial implementation

TODO

B.3 Phase 2: Distributed Infrastructure and Networking

TODO

Bibliography

- [1] Robin Coutelet. “Chronological vs. non-chronological backtracking in satisfiability modulo theories”. Unpublished master’s thesis. MA thesis. Liège, Belgique: Université de Liège, 2023. URL: <https://matheo.uliege.be/handle/2268.2/18041>.
- [2] Hans van Maaren Armin Biere Marijn Heule and Toby Walsch. “Handbook of Satisfiability”. In: (2008). URL: <https://www.cs.cornell.edu/~sabhar/chapters/ModelCounting-SAT-Handbook-prelim.pdf>.
- [3] Zhenghang Xu, Minghao Yin, and Jean Marie Lagniez. “An Embarrassingly Parallel Model Counter”. In: *Proceedings of the 22nd International Conference on Principles of Knowledge Representation and Reasoning*. Oct. 2025, pp. 670–681. DOI: [10.24963/kr.2025/65](https://doi.org/10.24963/kr.2025/65). URL: <https://doi.org/10.24963/kr.2025/65>.
- [4] Mate Soos and Kuldeep S. Meel. “Engineering an Efficient Probabilistic Exact Model Counter”. In: *Proceedings of the International Conference on Computer Aided Verification (CAV)*. 2025.
- [5] Shubham Sharma et al. “GANAK: A Scalable Probabilistic Exact Model Counter”. In: *Proceedings of International Joint Conference on Artificial Intelligence (IJCAI)*. 2019.
- [6] Armin Biere et al. “CaDiCaL 2.0”. In: *Computer Aided Verification - 36th International Conference, CAV 2024, Montreal, QC, Canada, July 24-27, 2024, Proceedings, Part I*. Ed. by Arie Gurfinkel and Vijay Ganesh. Vol. 14681. Lecture Notes in Computer Science. Springer, 2024, pp. 133–152. DOI: [10.1007/978-3-031-65627-9_7](https://doi.org/10.1007/978-3-031-65627-9_7).